Jonathan A. Diller

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Research Vision

My vision is to transform resource awareness into a foundational driver of autonomous decision-making in multi-agent robotics. I aim to pioneer algorithms that treat physical, computational, and communicative constraints not as operational hurdles, but as active guides for real-time reasoning and decentralized coordination. By bridging high-level intelligence with low-level resource management, I develop algorithms that enable resilient multi-agent coordination under real-world uncertainty. I validate these theoretical frameworks through physical deployments in high-stakes domains, including precision agriculture, defence and security, and normative AI alignment.

Education

Doctor of Philosophy in Robotics **August 2025**
Colorado School of Mines
Advisor: Dr. Qi Han
Thesis: *"Robust Robot Teams; Making Planning for Multi-Agent Systems Reflective"*

Master of Science in Computer Science **May 2022**
Colorado School of Mines

- GPA: *4.0/4.0*

Bachelor of Science in Computer Science **May 2020**
Pennsylvania State University, Harrisburg

- Minor in Mechatronics Technology
- Minor in Mathematics
- Graduated summa cum laude
- GPA: *3.99/4.0*

Research Experience

Postdoctoral Researcher *(Mentor: Dr. Vijay Kumar)* **September 2025 - Present**
GRASP Laboratory, University of Pennsylvania *Philadelphia, PA*

- Decision making for robot teams using semantic knowledge
- Adaptive information gathering for agricultural robots
- Evaluating embodied AI through the lens of societal norms

Research Assistant *(Advisor: Dr. Qi Han)* **July 2020 - August 2025**
Pervasive Computing Systems Group, Colorado School of Mines *Golden, CO*

- Studying and implementing multi-robot projects with focus on communication and energy-aware planning
- Writing proposals (contributed to two white papers)

Journeyman Fellow *(Mentor: Dr. J. Humann and J. Dotterweich)* **June 2024 - August 2024**
Army Research Laboratory *Golden, CO (remote)*

- Researched and developed planning algorithms for mixed drone-UGV teams.

Graduate Research Intern *(Mentor: Dr. John Rogers)* **May 2023 - August 2023**
Army Research Laboratory *Adelphi, MD*

- Researched the topics of task allocation and communication mapping for robot teams.

Research Scholar *(Advisor: Dr. Peter Idowu)* **June 2019 - July 2019**
Penn State Drawdown REU Program *Middletown, PA*

- Designed and evaluated algorithms for controlling microgrids using PLCs.

Research Assistant *(Advisor: Dr. Javad Khazaei)* **March 2019 - June 2019**
Pennsylvania State University *Middletown, PA*

- Researched and developed microcontroller applications for use in renewable energy projects.

Publications

† Student mentees, acceptance rate listed when known

Under Review

[R5] **J. Diller**, S. Homsi, and Q. Han. "*Semper Gumby: Making Multi-Robot Tasking Flexible under Changing Model Uncertainty*." Under review for the International Conference on Autonomous Agents and Multiagent Systems (AAMAS).

[R4] **J. Diller**, F. Cladera, C.J. Taylor, and V. Kumar. "*DORA: Divergence-Oriented Data-Relay Algorithm for Partially Connected Robot Teams*." Under review for the IEEE International Conference on Robotics and Automation (ICRA).

[R3] B. Li†, J. Romero, **J. Diller**, V. Kumar, A. Ribeiro. "*Visual Navigation Transformer with Pose Attention*." Under review for the IEEE International Conference on Robotics and Automation (ICRA).

[R2] **J. Diller**, D. Barnes, R. Bogdanoff, R. Collier, R. Collins, K. Fieldhouse, Y. Gefen, C. Johnson, A. Kodali†, B. Kriel, V. Murali, J. Niehaus, M. Sukharev, J. VanPelt, A. Hoogs, V. Kumar, and A. Basharat. "*REBAR: Reference Ethical Benchmark for Autonomy Readiness*." Under review for the Springer Proceedings in Advanced Robotics.

[R1] J. Zhang†, Y. Wu†, G. Shi, **J. Diller**, G. Sukhatme, V. Kumar. "*EFLUX: Elastic Multi-Robot Formation Navigation and Adaptation with Agentic LLMs*." Under review for the IEEE Robotics and Automation Letters.

Journal Papers

[J4] **J. Diller**, Q. Han, and M. Desaulniers†. "*Holistic Framework for Multi-Drone Data Collection in IoT Systems*." Accepted to the ACM Transactions on Internet Technology. 2026.

[J3] **J. Diller**, J. Sigler† and Q. Han. "*Collaborative Path Planning of Energy-Sharing Drone-UGV Teams for Persistent Patrolling*." ACM Journal on Autonomous Transportation Systems. Volume 3, Issue 4, Article 21. December, 2026.

[J2] **J. Diller**, J. Rogers, Q. Han, and N. T. Dantam. "*Communication Mapping for Robot Teams*." IEEE Transactions on Field Robotics. Volume 2, pp. 288-307. 2025.

[J1] **J. Diller**, Q. Han. "*Energy-Aware Drone Path Finding with a Fixed-Trajectory Ground Vehicle*." ACM Journal on Autonomous Transportation Systems. Volume 2, Issue 4, Article 17. December, 2025.

Conference Papers

[C8] Z. Ravichandran†, **J. Diller**, F. Cladera, V. Murali, G. Pappas, V. Kumar. "*Actively Resolving Contextual Uncertainty for Underspecified Tasks in Natural Language*." Proceedings of the International Symposium of Robotics Research (ISRR), 2026.

[C7] **J. Diller**, Y. S. Teoh†, R. Byers†, Q. Han, J. G. Rogers, and N. T. Dantam. "*Failure-Aware Tasking for Teams of Drones*." Proceedings of the International Conference on Distributed Computing Systems (ICDCS), 2025. *Acceptance rate: 19.7%*.

[C6] **J. Diller**, Q. Han, R. Byers†, J. Dotterweich, and J. Humann. "*Hitchhiker's Guide to Patrolling: Path-Finding for Energy-Sharing Drone-UGV Teams*." Proceedings of the International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2025. *Acceptance rate: 24.5%*.

[C5] C. Wang, **J. Diller**, and Q. Han. "*LLM for Generating Simulation Inputs to Evaluate Path Planning Algorithms*." Proceedings of the International Conference on Machine Learning and Applications (ICMLA), 2024. *Acceptance rate: 24.3%.*

[C4] **J. Diller**, P. Hall†, and Q. Han. "*Holistic Path Planning for Multi-Drone Data Collection*." Proceedings of the International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT), 2023. *Acceptance rate: 38.4% (short paper)*.

[C3] **J. Diller** and Q. Han. "*Energy-Aware UAV Path Planning with Adaptive Speed*." Proceedings of the International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2023. *Acceptance rate: 23.3%*.

[C2] **J. Diller**, P. Idowu, J. Khazaei, "*Load-Leveling Trainer for Demand Side Management on a 45kW Cyber-Physical Microgrid*," Proceedings of the Texas Power and Energy Conference (TPEC), 2020

[C1] **J. Diller**, B. Trussell, J. Khazaei, P. Idowu, "*Hardware Development of a Three-Phase 3.5 kW SiC Converter with Sinusoidal PWM*," Proceedings of the Texas Power and Energy Conference (TPEC), 2020

Workshop Papers

[W8] J. Zhang†, Y. Wu†, G. Shi, **J. Diller**, G. Sukhatme, V. Kumar. . "*Multi-Robot Formation Coordination with Verified Agentic LLM Planning*." Invited for oral presentation at the Workshop on Foundation Models in Multi-Robot Systems, 2026. 🏆 *Best Paper Aware*.

[W7] M. Gellar†, **J. Diller**, F. Cladera, K. Huang, S. Rahman Rahmani, S. Carpin, V. Kumar. "*Design and Control of an Autonomous Soil pH Sensing System*." Presented at the From Innovation to Adoption: Hardware, AI, Policy, and Funding Challenges in Agricultural Robotics Workshop, 2026.

[W6] A.K. Srivastava†, **J. Diller**, V. Kumar, P. Dames. "*Spatial-Semantic Uncertainty in VLM-Based Target Search: Balancing Exploration and Identification*." Invited for oral presentation at the Rethinking Uncertainty for Modern Robotics Paradigms Workshop, 2026.

[W5] **J. Diller**, F. Cladera, C.J. Taylor, and V. Kumar. "*LMPath: Language-Mediated Priors and Path Generation for Aerial Exploration*." Presented at the AI-Driven Safe Aerial Robotics Workshop, 2026.

[W4] **J. Diller**, D. Barnes, R. Bogdanoff, R. Collier, R. Collins, K. Fieldhouse, Y. Gefen, C. Johnson, A. Kodali†, B. Kriel, V. Murali, J. Niehaus, M. Sukharev, J. VanPelt, A. Hoogs, V. Kumar, and A. Basharat. "*REBAR: Reference Ethical Benchmark for Autonomy Readiness*." Presented at the 2nd Workshop on Robot Ethics – Ethical, Legal and User Perspectives in Robotics and Automation (WOROBET), 2026.

[W3] **J. Diller**, N. Dantam, J. Rogers, and Q. Han. "*Communication Jamming-Aware Robot Path Adaptation*." In proceedings of the International Workshop on Distributed Collective Intelligence (DISCOLI), 2023.

[W2] P. Hall†, **J. Diller**, A. Moon†, and Q. Han. "*DroNS-3: Framework for Realistic Drone and Networking Simulators*." In proceedings of the Workshop on Micro Aerial Vehicle Networks, Systems, and Applications (DroNet), 2023.

[W1] **J. Diller**, P. Hall†, C. Schanker†, K. Ung†, P. Belous†, P. Russell†, and Q. Han. "*ICCSwarm: A framework for Integrated Communication and Control in UAV Swarms*." In proceedings of the Workshop on Micro Aerial Vehicle Networks, Systems, and Applications (DroNet), 2022.

Extended Abstracts & Demonstrations

[A4] M. Hatch†, C. Fellinge†, **J. Diller**, and Q. Han. "*ROARQuad: Robust, Open Academic Research Quadcopter*." Accepted for demonstration at the International Conference on Mobile Systems, Applications, and Services (Mobisys), 2025.

[A3] **J. Diller**. "*Self-Aware High-Level Planning for Robot Teams*." Presented at the Cyber-Physical Systems Rising Stars Workshop, 2024. *Acceptance rate: 16.4%*.

[A2] **J. Diller**. "*Planning and Coordination for Unmanned Aerial Vehicles*." Presented at the International Conference on Autonomous Agents and Multiagent Systems' Doctoral Consortium, 2023.

[A1] **J. Diller**, Q. Han, C. Dreyer, F. Rossi, S. Bandopadhyay, J.P. de la Croix, A. Rahmani, P. Clark. "*Integrated Communication and Controls for Swarms of Small Satellites*." Presented at the Inter-Planetary Small Satellite Conference, 2021.

Awards & Recognitions

- Best Paper Award - FM-MRS Workshop **Sept, 2026**
- Rath Research Award Nominee, *Colorado School of Mines* **Dec, 2025**
- 2024 Cyber-Physical Systems Rising Star **May, 2024**
- Best PhD Poster, C-MAPP, *Colorado School of Mines* **Jan, 2024**
- Best PhD Poster, C-MAPP, *Colorado School of Mines* **Jan, 2023**
- Best Elevator Pitch, C-MAPP, *Colorado School of Mines* **Feb, 2022**
- Graduation Student Marshal for School of Sci., Eng. and Tech. **May, 2020**
- Computer Science Outstanding Student, *Pennsylvania State University* **Apr, 2020**
- Evan Pugh Scholar Award - Senior, *Pennsylvania State University* **Apr, 2019**
- Evan Pugh Scholar Award - Junior, *Pennsylvania State University* **Apr, 2018**
- President's Freshman Award, *Pennsylvania State University* **Apr, 2017**
- Dean's List, *Pennsylvania State University* **Spring 2016 - Spring 2020**

Teaching Experience

Primary Instructor **August 2024 – May 2025**
Colorado School of Mines *Golden, CO*
- CSCI 128: Computer Science for STEM (Fall 2024, Spring 2025)
- Typical class size: 50-70 students

College Teaching Certificate Program **August 2024 – May 2025**
Colorado School of Mines, Trefny Center *Golden, CO*
- Studied teaching pedagogy and learner-centered teaching practices
- Designed an upper-level undergraduate course to introduce students to robot autonomy

Teaching Assistant **August 2020 – December 2022**
Colorado School of Mines *Golden, CO*
- CSCI 565: Distributed Systems (Fall 2022)
- CSCI 406: Algorithms (Spr 2021)
- CSCI 261: Programming Concepts (Fall 2020)

Peer Tutor **August 2017 – May 2020**
Pennsylvania State University *Middletown, PA*
- Tutor students in Computer Science, Mathematics, Physics and Engineering classes.
- Earned CRLA Certified Tutor, Level I Certification.

Mentoring Experience

Graduate Students *degree (year conferred)*

University of Pennsylvania

Alkesh Kumar Srivastava	*PhD in Mechanical Engineering*
Beiming Li	*PhD in Electrical and Systems Engineering*
Samarth Chopra	*PhD in Electrical and Systems Engineering*
Yu-Hsuan Lin	*MSE in Robotics*
Omar Hossain	*MSE in Robotics*
Mark Geller	*MSE in Robotics*
Ankit Prabhu (*Research Staff*)	*MSE in Robotics (2023)*
Anuriha Kodali	*MSE in Robotics (2026)*
Jinyuan Zhang	*MSE in Robotics (2026)*

Colorado School of Mines

Matthew Desaulniers	*PhD in Robotics*
Robert Byers	*MS in Computer Science*
Yee Shen Teoh	*MS in Robotics (2025)*
Priestly Barigala	*MS in Robotics (2024)*

Undergraduate Students

University of Pennsylvania

Gavin Wong	*BSE in Computer Science*
Kelly Zhang	*BSE in Computer Science*
Kim Huang	*BSE in Electrical and Systems Engineering*

Colorado School of Mines

Cody Fellinge*	*BS in Electrical Engineering (2025)*
Matthew Hatch*	*BS in Electrical Engineering (2025)*
Jack Sigler	*BS in Computer Science (2025)*
Ava Moon	*BS in Computer Science*
Corey Schanker	*Combined BS in CS + EE*
Peter Hall*	*BS in Computer Science (2023)*
Kristen Ung*	*BS in Electrical Engineering (2022)*
Philip Belous	*BS in Computer Science (2022)*
Zachary Smeton*	*BS in Computer Science (2021)*

* Graduated with Undergraduate Research Scholar Distinction

Industry Experience

Robert Bosch GmbH **May 2018 – July 2020**

Embedded Software Developer & Test Intern *Lancaster, PA*

- Develop firmware for embedded real-time environments.
- New product prototyping.
- Design and implement automated tests for moving cameras.

- Write documentation for internal procedures.

United States Marine Corps **November 2010 - November 2015**
KC-130J Crewmaster, Plane Captain *San Diego, CA & Okinawa, Japan*
- Last held rank/pay grade: Sergeant/E-5
- Supervised small teams in pre and post flight inspections on C-130J aircraft.

Service

Technical Reviewer
- International Conference on Autonomous Agents and Multiagent Systems
- IEEE Robotics and Automation Letters
- Pervasive and Mobile Computing Journal
- ACM Journal on Autonomous Transportation Systems
- IEEE International Conference on Pervasive Computing and Communications
- IEEE International Conference on Robotics and Automation
- IEEE/RSJ International Conference on Intelligent Robots and Systems
- ACM/IEEE International Conference on Cyber-Physical Systems
- International Conference on Unmanned Aircraft Systems
- IEEE Internet of Things Journal
- Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies
- IEEE International Conference on Robot and Human Interactive Communication
- IEEE Transactions on Field Robotics

Committees
- IROS Workshop on Tightly Coupled Physical Interaction and Collaboration in Multi-Robot Systems *2026*
- Graduate Research and Discovery Symposium, Program Chair *2025*
- Graduate Research and Discovery Symposium, Poster Chair *2024*
- Mines Research Integrity and Security Committee, Grad. Rep. *Fall 2023 - Spring 2024*
- Mines Research Council, Graduate Representative *Spring 2023*
- CS@Mines Research Committee, Student Advocate *Spring 2022 - Spring 2023*

Other Service
- Graduate Student Government, CS Representative *Fall 2022 - Spring 2024*
- Graduate Student Seminar Organizer:
  - 6 seminars in 2023
  - 4 seminars and 1 workshop in 2024
  - 2 seminars in 2025